

SINGULAR-VALUE DECOMPOSITION

AP

BACKGROUND

SPECTRAL METHODS: TRADITIONAL

Eigenpairs

$$A\vec{e} = \lambda\vec{e}$$

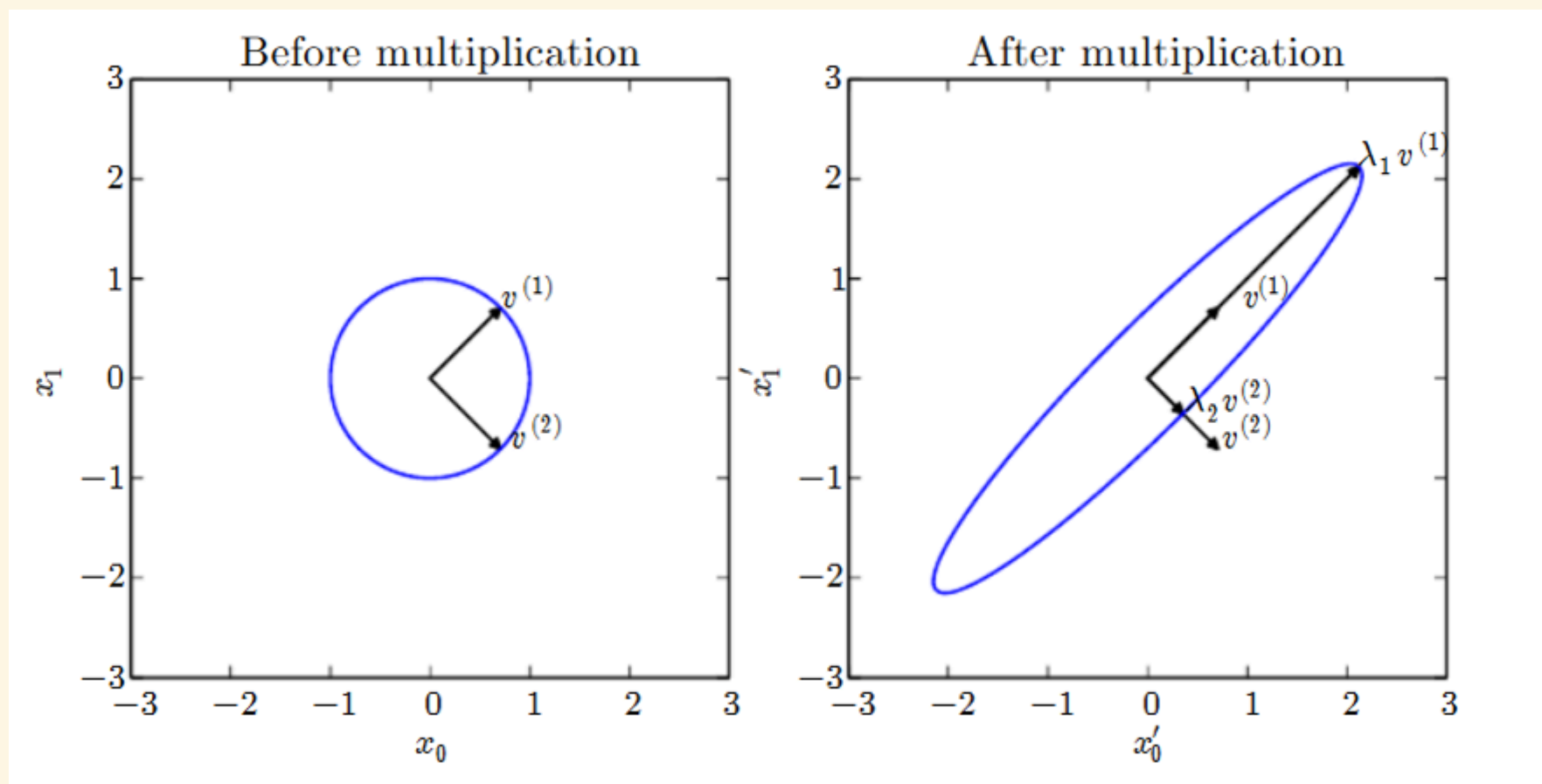


Figure 2.3: An example of the effect of eigenvectors and eigenvalues. Here, we have a matrix $\mathbf{A}$ with two orthonormal eigenvectors, $\mathbf{v}^{(1)}$ with eigenvalue λ_1 and $\mathbf{v}^{(2)}$ with eigenvalue λ_2 . (Left) We plot the set of all unit vectors $\mathbf{u} \in \mathbb{R}^2$ as a unit circle. (Right) We plot the set of all points $\mathbf{A}\mathbf{u}$. By observing the way that $\mathbf{A}$ distorts the unit circle, we can see that it scales space in direction $\mathbf{v}^{(i)}$ by λ_i .

EIGENDECOMPOSITION:

$$M = Q\Lambda Q^{-1}$$

M is interpreted as the combination of three specific transformations

Computing its inverse, M^{-1} , is now much simpler

THE IMPORTANCE OF MATRIX INVERSION

M. inversion solves linear systems of linear equations

$$M\vec{x} = \vec{v}$$

When M and $\vec{v}$ are know, we find $\vec{x}$

$$M^{-1}M\vec{x} = M^{-1}\vec{v}$$

$$I\vec{x} = M^{-1}\vec{v}$$

M. INVERSION UNDERPINS REGRESSION

The solution $\vec{x} = M^{-1}\vec{v}$ gives us the regression coefficients that define the linear regression model.

To estimate a new datapoint, say, $[a, b]$, just plug in the solution:

$$\hat{y} = [a, b] \cdot \vec{x}$$

Regression seen as solving the underlying linear system by matrix inversion

PRACTICAL ASPECTS

Textbook inversion takes $\Theta(n^3)$ operations: unfeasible

Advanced a. in Numpy run in $\Theta(n^2)$ (but check for numeric issues)

Compute M^{-1} once then store and re-use it for different $\vec{v}$

SQUARE MATRIX INVERSION

Thanks to properties of the Q matrix, where each column is an e-vector, inversion is simplified

$$M_{(m \times m)} = Q \Lambda Q^{-1}$$

$$M^{-1} = (Q \Lambda Q^{-1})^{-1}$$

inverse factor order: $M^{-1} = (Q^{-1})^{-1} \Lambda^{-1} Q^{-1}$

$$M^{-1} = Q \Lambda^{-1} Q^{-1}$$

where Λ^{-1} is obtained by simply substituting each λ_i on the diagonal with $\frac{1}{\lambda_i}$.

SINGULAR-VALUE DECOMPOSITION

For rectangular matrices $A_{(m \times n)}$ eigendecomposition is not defined

Instead, SVD provides a similar decomposition of the data matrix

The procedure is more complex

Until 1955, no general procedure for inverting rectangular m. was available

THE DATA MATRIX

$$A_{(m \times n)}$$

There are m points (rows) inside a *space* of n dimensions (columns)

Each point is represented by an m -dimensional *row* vector

Each dimension is represented by an n -dimensional vector

By multiplying a rectangular matrix by its transpose we obtain a square matrix that represents an 'internal' relationship:

$$M_{(m \times m)} = A_{(m \times n)} \times A_{(n \times m)}^T$$

$$N_{(n \times n)} = A_{(n \times m)}^T \times A_{(m \times n)}$$

Let's extract their respective eigenpairs.

NEW MATRIX: U

$U_{(m \times m)}$: columns made up of eigenvectors of

$$M_{(m \times m)} = A_{(m \times n)} \times A_{(n \times m)}^T.$$

Notation: let $\vec{U}_i$ be the i-th column of matrix U .

The e-vectors that make up U are always orthogonal with each other:

$$\vec{U}_i^T \cdot \vec{U}_j = 0 \quad (i \neq j).$$

this will simplify computation.

However, some further orthogonal column will have to be introduced with external methods as only $\min(m, n)$ non-zero eigenpairs can be obtained.

NEW MATRIX: V

$V_{(n \times n)}$: columns made up of eigenvectors of $N_{(n \times n)}$

Again, e-vectors are orthogonal to each other: $\vec{V}_i^T \cdot \vec{V}_j = 0 \ (i \neq j)$

NEW MATRIX: D (OR Σ)

dispose the eigenvalues of N on the main diagonal of a rectangular m . that will be 0 everywhere else

$D_{(n \times m)}$ where $D_{ii} = \sigma_i$ are the singular values

$\sigma_i = \sqrt{\lambda_i}$ the i -th e-value of $N = A^T A$

(it can also be constructed with $M = AA^T$)

FINALLY...

$$A_{(m \times n)} = U_{(m \times m)} D_{(m \times n)} V_{(n \times n)}^T$$

- U is a orthogonal m. of *left-singular* (col.) vectors
- D is a diagonal matrix of *singular values*
- V is a orthogonal m. of *right-singular* (col.) vectors

Please see § 2.7 of [\[Goodfellow et al.\]](#)

CONSEQUENCES

SVD generalises eigen-decomposition:

- any real matrix has one
- even non-square matrices admit one

over-constrained systems (more rows than cols.) can now be solved

WORKED-OUT EXAMPLE

EXAMPLE

$$A_{(3 \times 2)} \vec{x} = \vec{v}$$

Focus on

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix}$$

Extract the eigenpairs: $\lambda_1 \approx 91, \vec{e}_1 = [0.15, 0.33, 0.52]$

Unfortunately, $\lambda_2 \approx \lambda_3 \approx 0$: orthogonal vectors are needed to make up our matrix U

(the third row is almost the sum of the first two, i.e., almost dependent)

$$A^T A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 35 & 44 \\ 44 & 56 \end{pmatrix}$$

The eigenvalues are $\lambda_1 = 90.54$ and $\lambda_2 = 0.26$

The singular values are $\sigma_1 = \sqrt{\lambda_1} = 9.54$ and $\sigma_2 = \sqrt{\lambda_2} = 0.51$

$$D = \begin{pmatrix} 9.53 & 0 \\ 0 & 0.51 \\ 0 & 0 \end{pmatrix}$$

Putting it all together:

$$A = \begin{pmatrix} 0.15 & 0.231 & 0.882 \\ 0.33 & 0.527 & 0.216 \\ 0.52 & 0.823 & -0.451 \end{pmatrix} \begin{pmatrix} 9.53 & 0 \\ 0 & 0.51 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0.62 & 0.79 \\ -0.79 & 0.62 \end{pmatrix}$$

Even though this formulation is far from the original and not easily interpreted, it has good properties.

STUDY PLAN

BACKGROUND STUDY

Ian Goodfellow, Yoshua Bengio and Aaron Courville: [Deep Learning](#), MIT Press, 2016.

available in HTML and PDF; it is *a refresher* of notation and properties: no examples and no exercises.

It can be read in the background.

- Phase 1: read §§ 2.1–2.7, then § 2.11.
- Phase 2: read §§ 2.8–2.10

MOORE-PENROSE PSEUDO-INVERSE

MOTIVATIONS

solve linear systems

$$A\vec{x} = \vec{v}$$

for non-square (rectangular) matrices:

- rows > columns: the problem is over-constrained (no solution?)
- rows < columns: the problem is over-parametrized (infinite sols.?)

REMINDER: THE IDEAL PROCEDURE

If A is square ($m=n$) and non-singular ($|A| \neq 0$) then

$$A\vec{x} = \vec{v}$$

$$A^{-1}A\vec{x} = A^{-1}\vec{v}$$

$$I\vec{x} = A^{-1}\vec{v}$$

Compute A^{-1} once, run for different values of $\vec{v}$

THE PSEUDO-INVERSE

Given $A_{m \times n}$, Penrose seeks a matrix A^+ that would work the same as the left inverse:

$$A^+ A \approx I$$

Idea: strengthen up the main diagonal before inversion:

$$A^+ = \lim_{\alpha \rightarrow 0} (A^T A + \alpha I)^{-1} A^T$$

Thanks to A^+ , over-constrained linear systems can now be solved (w. approximation): a new, general method for solving Phys/Eng. problems

SVD LEADS TO APPROX. INVERSION

for the decomposition

$$A = UDV^T$$

$$A^+ = VD^+U^T$$

where D^+ , such that $D^+D = I$ is easy to calculate: if

$$\mathbf{D} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

then set

$$\mathbf{D}^+ = \begin{pmatrix} \frac{1}{a} & 0 \\ 0 & \frac{1}{b} \end{pmatrix}$$

Does $A^+ A \approx I$?

Yes, because U and V are s. t. $U^T U = V V^T = I$.

$$V D^+ U^T \cdot U D V^T =$$

$$V D^+ I D V^T =$$

$$V D^+ D V^T =$$

$$V I V^T = V V^T = I$$